

Anu, Ryan and Will
3-15-26
Stat Comps

Progress Report

Data Selection:

Initially, we planned to use the methodology as well as data from the Guyll 2023 article. However, due to issues in statistical reasoning as well as the data missing key explanatory variables, we pivoted to using the data from Monson et. al 2022 but following the methodology from Busey et. al 2023. This data contains both observations from bullets and cartridge casings. We decided to filter to only include the cartridge casings section in order to be consistent with the data Guyll used. Additionally, there were redundant variables as well as ones that were irrelevant to our study, so we made the decision to filter those out as well. This can be found in `clean_data.csv` in our github repository.

Analyzed Literature:

Aside from the aforementioned Guyll and Monson data sets, which provide key insight into methodology and data, we also plan on using work from Busey et. al 2023 and Rosenblum et. al. The Busey paper provides us with a relevant modeling approach that is often used in forensics, the ordinal probit. We plan on using the ordinal probit to classify decisions that examiners make with regards to the identification of a gun to a crime scene as well as gain insight into how difficult specific guns are to make correct decisions. Rosenblum provides a response and critique to Guyll's reasoning, citing the fact that Guyll's conclusions were too skewed towards the prosecution, which was conducive to inflated convictions. We aim to synthesize these four pieces of literature into a cohesive piece that blends the concepts and data put forth.

EDA:

We have also done EDA on the filtered Monson data. This EDA can be found in the `eda.qmd` file in our github repository. The EDA includes visualizations of the decisions made by examiners for both mated and non mated pairs of guns as well as the accuracy of decisions made on the examiner level. We looked at these decisions on a per-gun basis as well as on a per-gun-pair basis. These provide valuable insight into the data as we are able to identify outliers on the gun as well as examiner level that could skew results in one way or another.

Modeling:

We started off trying to replicate Busey's JAGS sampling approach but that proved difficult, so we took advantage of the ordinal probit implementation from *rstanarm*. We fit the ordinal probit on two levels of the data, the gun level both for mated and non-mated comparisons. In the future, we might take into account individual examiners' identities, but following from Busey, the likelihood ratio is calculated based on the individual latent distributions from the posterior of the fitted model. After obtaining estimated coefficients from the chain, we looked at the posterior predictive check to analyze model fit, as well as convergence. We then looked at the posterior

distributions of the coefficients themselves and, treating them as probability density functions, we added and normalized them like Busey did. Currently, we are in the process of designing simulations as a proof of concept that we can recover latents from data.

Planned Future Directions:

Next term, we plan on interpreting our results in context and establishing priors and likelihood ratios based on our modeling results. Additionally, we will ensure that the literature and methodology builds off each other and is a cohesive unit as opposed to four disjoint sections. We are still working on exactly what the direction and deliverable aspect of our final report will be.